

SUPPORTLYTICS

Optimizing IT Support Team Performance Using Advanced Analytics

PROJECT DOCUMENTATION

DATA ANALYTICS • PYTHON • GOOGLE COLAB • POWER BI

Prepared By

MATHETI JAGANNATH REDDY

Mentor

Shanmukha Priya

Internship Program

Infosys Springboard Internship

Project Repository

<https://github.com/springboardmentor1210a/Supportlytics/tree/JagannathReddy/Supportlytics>

Project Notebook

Google Colab analysis notebook

Project Dataset Snapshot: 8,469 support-ticket records • 19 fields • 6 Power BI dashboards • 32.70% records with a populated Resolution Hours value

This document presents the complete analytical workflow, data preparation, feature engineering, exploratory analysis, Power BI dashboards, cluster analysis, service-quality analysis, recommendations and final submission structure.

1. Project Overview

Supportlytics is an IT support analytics project designed to transform ticket-level support records into structured operational insights. The project examines workload, ticket status, priority, ticket type, customer characteristics, products, service channels, satisfaction, recurring issue groups and geographic/category patterns.

The analytical workflow combines Python and pandas for data preparation, feature creation and exploratory analysis with Power BI for interactive management reporting. The final documentation follows the project flow from data understanding through dashboard reporting and recommendations.

2. Project Objectives

- Analyze historical IT support-ticket data and establish a reliable analytical baseline.
- Understand the dataset schema, data types, categorical distributions and missing values.
- Clean inconsistent, incomplete and incorrectly typed fields before visualization.
- Prepare text fields for token and recurring-issue analysis.
- Create analytical features such as Resolution Duration and Priority Score.
- Compare workload and performance by ticket type, priority, channel, team, product and geographic dimensions.
- Apply clustering and similarity analysis to group related support issues.
- Identify high-priority and unresolved work that may require escalation.
- Evaluate customer satisfaction and service quality from multiple perspectives.
- Build six interactive Power BI dashboards and convert analytical outputs into management recommendations.

3. Business Problem

IT support teams can receive large numbers of tickets across different issue types, priorities, channels and products. A ticket list alone does not reveal whether workload is balanced, whether recurring issues are consuming resources, or whether high-severity work is being resolved effectively. Supportlytics addresses this gap by organizing raw ticket records into measurable KPIs, comparative visuals, cluster-level patterns and management-oriented dashboards.

4. Technology Stack

Technology	Role in the project
Python	Data cleaning, preprocessing, feature engineering and analysis
Pandas / NumPy	Data manipulation, validation and numerical processing
NLTK / WordCloud	Text normalization, token analysis and common-issue visualization
Matplotlib / Seaborn	Exploratory and analytical visualization
Google Colab	Notebook-based analysis and execution environment
Power BI	Interactive dashboards, KPIs, filtering and management reporting
Excel	Supplementary validation and checking
GitHub	Version control, repository organization and project submission

5. Dataset and Data Preparation

The cleaned Supportlytics dataset contains 8,469 records and 19 fields. The fields cover customer information, product information, ticket characteristics, service status, priority, channel, first-response information, resolution information and customer satisfaction.

5.1 Main Data Fields

Customer: Customer Age, Customer Gender, Customer Name and Customer Email.

Ticket: Ticket ID, Ticket Type, Ticket Subject, Ticket Description, Ticket Priority and Ticket Status.

Product: Product Purchased and Date of Purchase.

Service Quality: Customer Satisfaction Rating, First Response Time, First Response Hours and Resolution Hours.

Operations: Resolution, Ticket Channel and Time to Resolution.

Analysis: Cluster_ID, Cluster_Name and Similarity_Score are used in the clustered analytical layer.

Geographic: Country, Region, Latitude and Longitude are enriched attributes used for geographic visualization.

5.2 Data Quality and Cleaning

- Inspect dataset dimensions, sample records, data types and descriptive statistics.
- Check missing values in date, response, resolution and satisfaction fields.
- Normalize text fields before token and frequency analysis.
- Convert date/time fields to consistent datetime representations.
- Validate numerical fields and identify values that require interpretation before benchmarking.
- Check duplicate records and inconsistent categorical values.
- Prepare the cleaned analytical dataset before connecting the data to Power BI.

5.3 Important Data-Quality Note

Two data-quality considerations are retained: (1) Country, Region, Latitude and Longitude were enriched for geographic visualization; and (2) some calculated time values require validation before they are used as operational benchmarks. These fields should be treated as analytical/demo attributes unless validated against source-system records.

6. Analytical Methodology

The project follows a sequential analytical workflow so that dashboard outputs are supported by prepared and examined data. Each stage contributes a specific analytical purpose and feeds the next stage.

Step 1 — Data Collection & Understanding

Load the support-ticket CSV and inspect its schema, types, dimensions and distributions.

Step 2 — Data Preprocessing

Handle incomplete, duplicate, inconsistent or incorrectly typed values and prepare analysis-ready fields.

Step 3 — Feature Engineering

Create response/resolution and priority-related analytical measures.

Step 4 — Exploratory Data Analysis

Study ticket volume, ticket type, priority, status, products, channels and customer characteristics.

Step 5 — Text Analysis

Normalize ticket descriptions, tokenize text and identify frequent terms and recurring issue themes.

Step 6 — Clustering & Similarity

Group related tickets and compare cluster size, similarity and performance.

Step 7 — Performance Analysis

Compare response/resolution behavior and examine unresolved workload.

Step 8 — Customer Analysis

Evaluate satisfaction across ticket type, priority, channel, product and customer characteristics.

Step 9 — Geographic Analysis

Compare enriched country and region dimensions while retaining the geographic data-quality caveat.

Step 10 — Power BI Reporting

Translate analytical results into six interactive dashboards for operational and management use.

7. Feature Engineering

Resolution Duration measures the elapsed time associated with ticket resolution and provides a common basis for comparing service performance. Priority Score provides a numerical representation of priority severity so the workload can be aggregated and compared across tickets, categories, teams and issue clusters.

These engineered fields support the later Power BI measures and cluster-level comparisons. Because the source documentation flags some calculated time values for validation, the report uses them primarily for comparative analysis rather than claiming that every calculated duration is a validated operational SLA benchmark.

8. Exploratory Data Analysis

Exploratory analysis establishes the workload profile before dashboard interpretation. The following observations are derived from the cleaned project dataset and are used to explain why the dashboard dimensions were selected.

8.1 Ticket Status

- Open tickets: 2,819.
- Closed tickets: 2,769.
- Pending Customer Response: 2,881.
- The three-state distribution makes status a central operational dimension for identifying outstanding workload.

8.2 Ticket Priority

- Critical: 2,129 (25.1%).
- High: 2,085 (24.6%).
- Medium: 2,192 (25.9%).
- Low: 2,063 (24.4%).
- The near-balanced priority mix supports the use of priority as a comparison dimension rather than relying only on total volume.

8.3 Ticket Type and Channel

- Ticket types include Refund request (1,752), Technical issue (1,747), Cancellation request (1,695), Product inquiry (1,641) and Billing inquiry (1,634).
- Channels are distributed across Email (2,143), Phone (2,132), Social media (2,121) and Chat (2,073).

8.4 Customer Profile and Satisfaction

- Average customer age: 44.0 years; observed range: 18–70.
- Positive satisfaction ratings are available for 2,769 records, with an average of 2.99.
- Zero/unrated satisfaction values occur in 5,700 records, so satisfaction should be interpreted with awareness of rating coverage.

9. Power BI Dashboard Suite

The final Power BI report is organized into six complementary dashboard pages. The sequence moves from overall operational status to resolution performance, customer/product behavior, satisfaction, issue clustering and geographic/category insights.

Dashboard	Primary analytical purpose
1. IT Support Overview	Provides the management-level snapshot of support workload, status, priority and overall service performance.
2. Resolution & Performance	Focuses on resolution efficiency, outstanding workload and escalation-oriented analysis.
3. Customer & Product Insights	Connects customer characteristics, product demand and ticket-type workload.
4. Customer Satisfaction & Service Quality	Evaluates service quality through customer satisfaction, ticket type, priority, channel, product and demographic views.
5. Cluster & Similarity Analysis	Groups related support issues and compares cluster volume, similarity and resolution behavior.
6. Geographic & Category Insights	Compares support workload and performance across enriched countries and regions while retaining category-level context.

Dashboard Design Principles

- Use KPI cards for immediate management-level status.
- Use categorical charts for workload and composition comparisons.
- Use trend and distribution visuals for operational behavior.
- Use detailed tables where escalation or ticket-level follow-up is required.
- Use cluster and geographic visuals for higher-level pattern discovery.
- Keep conclusions tied to the actual fields and measures used in Power BI.

The next six sections document each Power BI page with its dashboard image, analytical coverage, and management interpretation.

Dashboard 1 – IT Support Overview

Provides the management-level snapshot of support workload, status, priority and overall service performance.

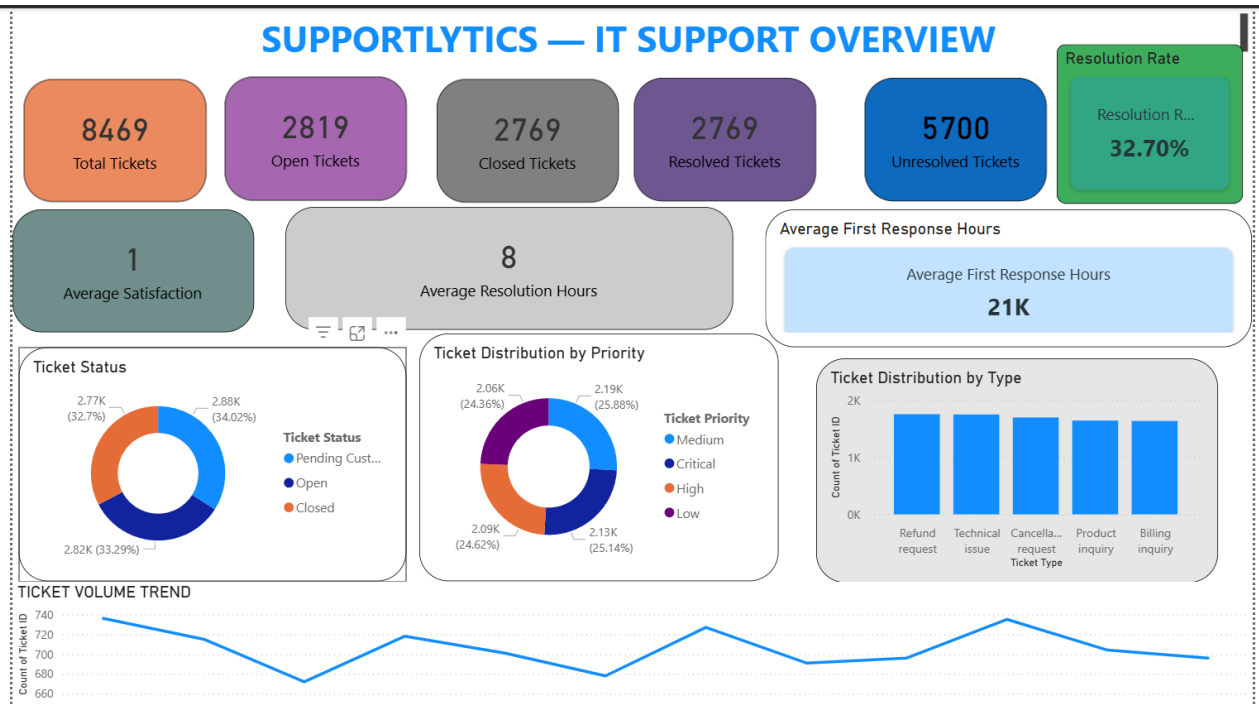


Figure 1. Supportlytics — IT Support Overview

Dashboard Data & Analytical Coverage

- Total ticket volume: 8,469 records across 19 analytical fields.
- Status distribution: 2,819 Open, 2,769 Closed and 2,881 Pending Customer Response.
- Priority distribution is relatively balanced: 2,129 Critical, 2,085 High, 2,192 Medium and 2,063 Low.
- Resolution coverage is approximately 32.70% based on non-null Resolution Hours.
- The page combines KPI cards with status, priority, ticket-type and volume-trend visuals so a manager can move from the overall workload to specific operational patterns.

Management Interpretation

This dashboard should be used as the first checkpoint during a support review. It quickly shows whether workload is increasing, whether unresolved/open work is accumulating, and how demand is distributed across priority and ticket type.

Recommended use: start-of-day operational review, weekly support meetings and management summaries.

Dashboard 2 – Resolution & Performance

Focuses on resolution efficiency, outstanding workload and escalation-oriented analysis.



Figure 2. Dashboard 2 — Resolution & Performance

Dashboard Data & Analytical Coverage

- 2,769 records contain a Resolution Hours value, forming the basis for the resolution-focused visuals.
- Resolution performance is compared by ticket priority and ticket type to identify groups that require closer operational attention.
- Channel workload is distributed across Email (2,143), Phone (2,132), Social media (2,121) and Chat (2,073).
- Priority-versus-status analysis helps identify high-severity work that remains outside a resolved state.
- The processed dataset contains resolution-time fields, but the project documentation notes that some calculated time values require validation before being used as operational benchmarks.

Management Interpretation

The key management use is escalation. High and Critical tickets should be reviewed together with their current status and resolution behavior rather than by volume alone.

Recommended use: escalation reviews, workload prioritization and investigation of slow-performing ticket groups.

Dashboard 3 – Customer & Product Insights

Connects customer characteristics, product demand and ticket-type workload.

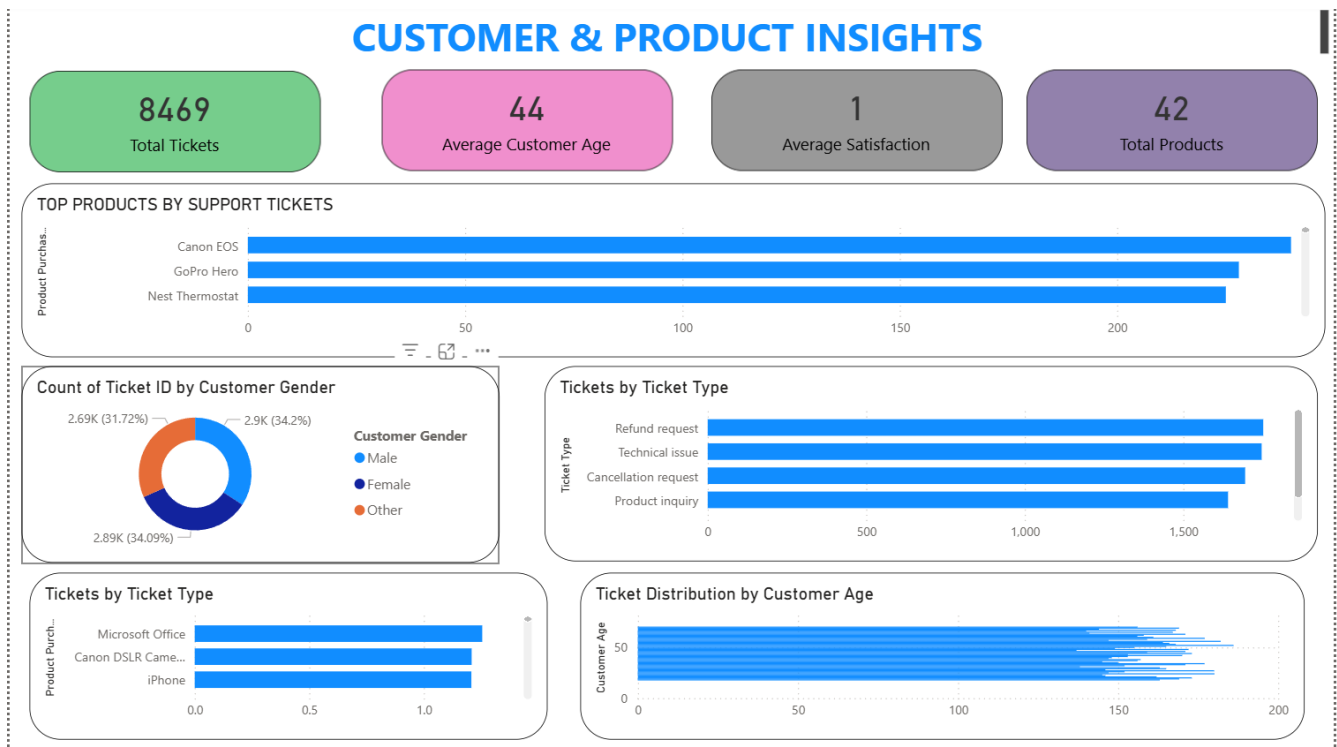


Figure 3. Dashboard 3 — Customer & Product Insights

Dashboard Data & Analytical Coverage

- Average customer age is approximately 44.0 years, with customer ages ranging from 18 to 70.
- Customer gender distribution: Male 2,896, Female 2,887, Other 2,686.
- The largest product-support volumes include Canon EOS (240), GoPro Hero (228) and Nest Thermostat (225) tickets.
- Ticket-type demand is led by Refund request (1,752), followed by Technical issue (1,747).
- The dashboard helps relate product demand to support workload and customer characteristics.

Management Interpretation

Product teams can use this view to identify products generating repeated support demand, while support managers can compare the demand profile with staffing and specialist coverage.

Recommended use: product-support planning, specialist allocation and identification of products generating repeated demand.

Dashboard 4 – Customer Satisfaction & Service Quality

Evaluates service quality through customer satisfaction, ticket type, priority, channel, product and demographic views.

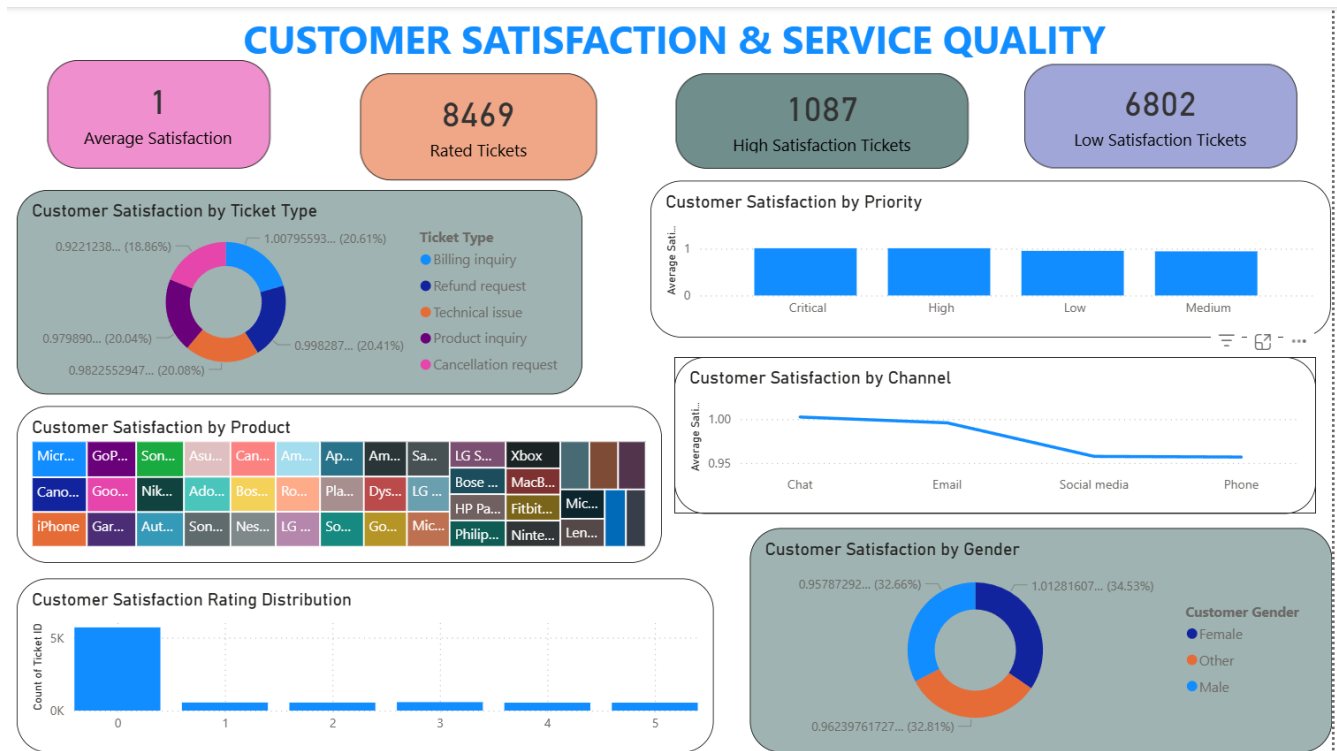


Figure 4. Dashboard 4 — Customer Satisfaction & Service Quality

Dashboard Data & Analytical Coverage

- The dataset contains 8,469 ticket records; 5,700 currently carry a zero/unrated satisfaction value.
- 2,769 records contain a positive satisfaction rating, with an average rated score of approximately 2.99 on the available scale.
- Rated-ticket distribution includes 1-star: 553, 2-star: 549, 3-star: 580, 4-star: 543, and 5-star: 544.
- Satisfaction is compared across ticket type, priority, channel, product and gender so low-performing service areas can be isolated.
- The dashboard separates satisfaction measurement from simple ticket volume, making it useful for service-quality discussions.

Management Interpretation

A useful management approach is to combine low satisfaction with ticket category, priority and resolution behavior. This can distinguish process problems from isolated customer-specific cases.

Recommended use: service-quality reviews, customer-experience analysis and investigation of low-rated support areas.

Dashboard 5 – Cluster & Similarity Analysis

Groups related support issues and compares cluster volume, similarity and resolution behavior.

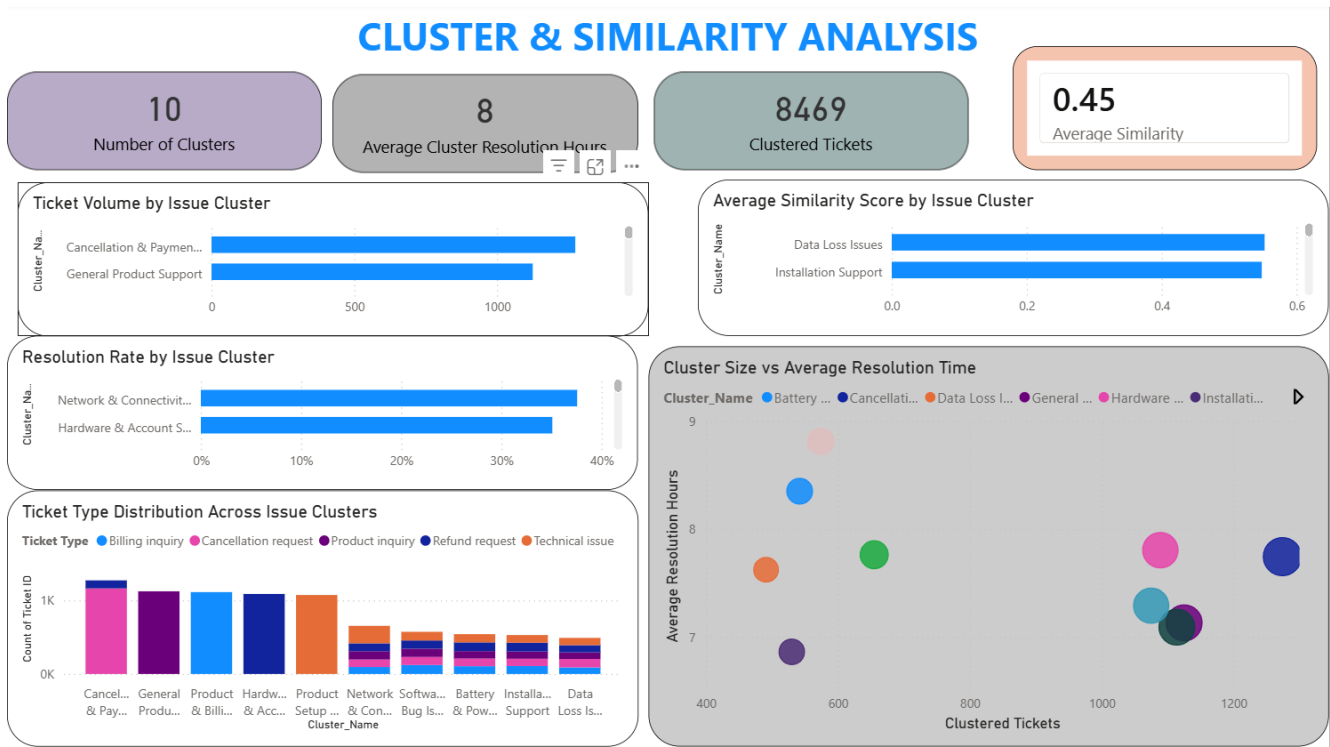


Figure 5. Dashboard 5 — Cluster & Similarity Analysis

Dashboard Data & Analytical Coverage

- The updated dashboard contains 10 issue clusters and uses Cluster_ID, Cluster_Name and Similarity_Score for comparison.
- The clustered view covers all 8,469 ticket records.
- Cluster volume identifies recurring issue families, while average similarity indicates how consistently tickets within a cluster resemble one another.
- Resolution rate by cluster highlights issue groups that may require specialized knowledge or workflow changes.
- Cluster size versus average resolution time provides a practical way to prioritize high-volume clusters with slower performance.

Management Interpretation

Clustering moves the project beyond simple categorical reporting. Instead of treating every ticket independently, managers can investigate recurring issue families and develop reusable troubleshooting guides, routing rules and knowledge-base articles.

Recommended use: recurring-issue identification, knowledge-base planning, routing and specialist workflow design.

Dashboard 6 – Geographic & Category Insights

Compares support workload and performance across enriched countries and regions while retaining category-level context.

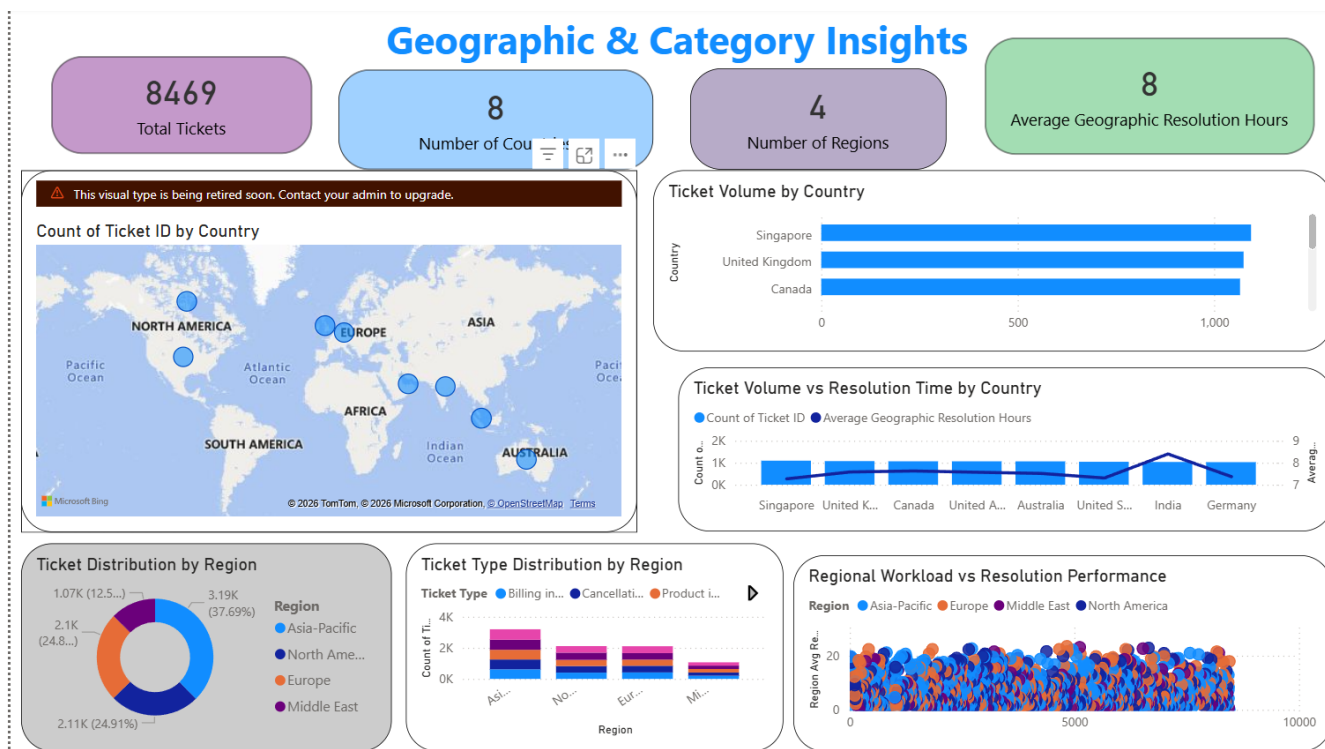


Figure 6. Dashboard 6 — Geographic & Category Insights

Dashboard Data & Analytical Coverage

- The dashboard provides country-level ticket concentration and regional workload comparisons.
- Ticket volume can be compared with resolution behavior to identify locations or regions that warrant additional operational review.
- Ticket type distribution by region helps reveal whether regional workloads differ in composition.
- The regional workload-versus-performance view supports resource allocation and process comparison.
- The geographic attributes were enriched for dashboard demonstration and should not be interpreted as verified original customer-location facts.

Management Interpretation

This page is most useful as a comparative management view. Regional differences can be investigated further before making staffing or process decisions, especially because the geographic fields are enriched rather than original source measurements.

Recommended use: regional comparison and workload/resource review, with the geographic enrichment caveat kept in mind.

10. KPI Framework and Measurement Logic

The dashboard suite uses KPIs to provide a consistent measurement layer across the six pages. KPIs should be interpreted together rather than in isolation because workload, resolution behavior and satisfaction can influence one another.

KPI	Purpose	Interpretation
Total Tickets	Overall support workload	Track volume and change over time.
Open / Unresolved Tickets	Outstanding workload	Higher values indicate more work requiring follow-up.
Resolution Rate	Completed-work proportion	Useful for comparing operational completion.
Average Resolution Hours	Resolution efficiency	Lower is generally preferable, subject to data validation.
Average First Response Hours	Initial responsiveness	Lower values indicate faster initial response.
Average Satisfaction	Customer experience	Higher rated values indicate stronger perceived service.
Average Priority Score	Workload severity	Helps compare severity across groups.
Average Similarity Score	Cluster consistency	Higher values indicate stronger similarity within issue groups.

11. Key Findings

- The project contains 8,469 ticket records, providing a substantial basis for comparative support analytics.
- Open and Pending Customer Response work together account for 5,700 records, making outstanding workload a major management dimension.
- Priority categories are relatively balanced, so high-priority workload should be evaluated using both volume and status.
- Ticket demand is spread across five major ticket types, allowing service teams to compare issue-family workload.
- Support channels are also relatively balanced, which makes channel-level comparison meaningful.
- Customer satisfaction coverage is incomplete because a large number of records have zero/unrated values; rated-ticket analysis is therefore more informative than using the raw overall average alone.
- Cluster analysis adds a recurring-issue perspective that is not available from basic ticket-type reporting.
- Geographic analysis provides a useful visualization layer but must be interpreted as enriched analytical data rather than verified original customer-location data.

12. Analytical Insights by Management Area

Workload Management

The status distribution shows that a substantial portion of the dataset remains outside the Closed state. This supports the need for workload queues, escalation rules and periodic backlog review rather than relying only on total ticket volume.

Priority Management

Critical and High tickets together represent nearly half of the dataset. A priority-based review should therefore focus on whether these tickets are progressing appropriately, not simply how many exist.

Channel Management

Email, Phone, Social media and Chat are all represented at similar levels. This makes channel comparison useful for identifying differences in response patterns, customer satisfaction and resolution behavior.

Product and Issue Management

Product-level and ticket-type views can identify recurring support demand. High-volume products or issue families can become candidates for better documentation, troubleshooting guides or specialist ownership.

Customer Experience

Satisfaction should be evaluated alongside ticket type, priority, channel and product. This helps distinguish isolated low ratings from systematic service-quality issues.

Recurring Issue Management

Cluster analysis provides an additional layer for identifying related problems. High-volume clusters with slower performance are particularly useful candidates for process improvement and knowledge-base development.

13. Limitations and Data-Quality Notes

- Geographic attributes were synthetically enriched because the original cleaned dataset did not contain country, region, latitude or longitude fields.
- Some calculated time-related values require validation before they are used as operational benchmarks or SLA claims.
- Dashboard metrics should always be interpreted according to the exact fields and Power BI measures used.
- Zero/unrated satisfaction values reduce the usefulness of a simple overall satisfaction average; rated-ticket coverage should be considered.
- The report is designed for analytical and management demonstration. Operational decisions should be validated against live support-system records.

14. Recommendations

1. Implement priority-based escalation

Create clear review rules for High and Critical tickets, especially when they remain Open or Pending Customer Response.

2. Build a recurring-issue knowledge base

Use frequent ticket categories, text terms and clusters to create troubleshooting guides, FAQs and reusable resolution steps.

3. Balance workload using evidence

Compare workload and performance by team, channel, ticket type and cluster before reallocating staff or specialist coverage.

4. Monitor resolution behavior continuously

Track resolution measures over time and investigate groups that repeatedly show slower performance, while respecting the data-quality caveat.

5. Connect satisfaction with operations

Investigate low satisfaction together with ticket type, priority, channel and product to identify systematic service-quality problems.

6. Use clustering for routing and specialization

Recurring issue clusters can support automated routing, specialist queues and targeted knowledge-base content.

7. Review geographic insights carefully

Use the geographic dashboard for comparative visualization, but validate enriched location attributes before making factual regional decisions.

Expected Operational Benefits

- Faster identification of high-priority outstanding work.
- Better visibility into recurring support problems.
- More structured workload allocation across teams and channels.
- Improved ability to connect service quality with operational behavior.
- A reusable reporting layer for management review and future data refreshes.

15. Final Deliverables

- Cleaned dataset — cleaned_support_tickets.csv.
- Clustered and geographically enriched analytical datasets.
- Python / Google Colab analysis notebook.
- Power BI PBIX dashboard.
- Six dashboard screenshots and visuals.
- Comprehensive project documentation.
- Final presentation slides.
- README.md and GitHub repository.

16. Recommended GitHub Structure

Supportlytics/ → README.md, data/, notebooks/, powerbi/, dashboard_screenshots/, visuals/, documentation/, presentation/ and LICENSE.

17. Execution Guide

- Clone or open the project repository.
- Open the analysis notebook in Google Colab, Jupyter or VS Code.
- Install the required Python libraries.
- Load the raw support-ticket CSV.
- Run the notebook cells sequentially.
- Generate or refresh the cleaned analytical dataset.
- Load the processed dataset into Power BI.
- Refresh the six dashboards and verify KPI calculations.
- Review insights, recommendations and dashboard screenshots.

Required Python Libraries

```
pip install pandas numpy matplotlib seaborn nltk wordcloud
```


18. Conclusion

Supportlytics provides an end-to-end analytics solution for understanding IT support operations. The project combines data cleaning, feature engineering, text analysis, exploratory visualization, clustering, similarity analysis, customer satisfaction analysis and Power BI reporting. The six-dashboard design turns a large ticket dataset into complementary management views covering workload, resolution performance, customer and product behavior, service quality, recurring issue groups and geographic/category patterns.

The strongest value of the project is the connection between descriptive reporting and operational decision-making. Managers can move from overall workload to priority, channel, product, satisfaction and recurring issue patterns, then use those findings to guide escalation, knowledge-base development, workload balancing and service-quality improvement.

19. Final Submission Checklist

- README updated and consistent with the Power BI report.
- Raw and cleaned datasets included.
- Notebook/code included and reproducible.
- Power BI PBIX included.
- All six dashboard screenshots included in the correct order.
- Project documentation included.
- Presentation slides included.
- Recommendations and limitations documented.
- GitHub repository organized and ready for mentor review.

Final documentation principle: dashboard screenshots, analytical descriptions and conclusions should remain consistent with the fields, calculations and data-quality notes used in the project.